



Savitribai Phule Pune University

Project
Stage –II
Report on

Live Surveillance with Actionable Intelligence

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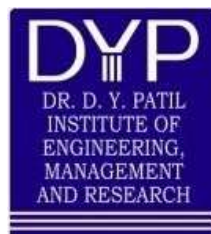
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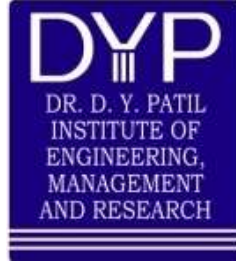
Under The Guidance of

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AKURDI, PUNE 44.**



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CERTIFICATE

This is to certify that **Mr. Arhant Sahuji, Mr. Tanay Shinde, Mr. Ritesh Kadam, and Ms. Ananya Vaishnav** of B.E. AI&DS have successfully completed Project Stage-II “**Live Surveillance with Actionable Intelligence**” towards the fulfillment of the requirements of the Degree of Engineering course under the Savitribai Phule Pune University, Pune, during the academic year 2025-2026.

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DECLARATION

We hereby declare that the entire project work entitled “**Live Surveillance with Actionable Intelligence**” is a project report of original work done by us, to the best of our knowledge and belief. No part of it has been submitted for any degree or diploma of any institution previously.

This project work is submitted to **Savitribai Phule Pune University, Pune, at the Dr. D. Y. Patil Institute of Engineering, Management, and Research, Akurdi, Pune**, during the academic year **2025-2026**.

Place:

Date:

Signature of students

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3. Ananya Vaishnav

4. Ritesh Kadam

ACKNOWLEDGEMENT

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Special thanks to our H.O.D. **Dr. Suvarna Patil** for her kind official support and encouragement. We are also thankful to our project coordinator **Mrs. Sneha Kanawade** for her valuable guidance. Finally, we would like to thank all our staff members of the AI & DS department who helped us directly or indirectly to complete this work successfully.

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2. Tanay Shinde
3. Ananya Vaishnav
4. Ritesh Kadam

ABSTRACT

The combination of advanced machine learning and AI capabilities associated with natural language processing (NLP) and technological developments in computer vision have led to the creation of smart video surveillance systems. Unlike conventional Closed Circuit Television (CCTV) and motion-detection-based surveillance devices, these systems are capable of understanding contextual information, reducing false alarms and requiring less human intervention. Researchers have explored incorporating vision-language models (VLMs) and Sentiment Analysis (SA) into video surveillance applications to improve contextual awareness. This review focuses on emerging techniques associated with image captioning models, including Salesforce's BLIP, used to generate natural language descriptions of real-time actions in video footage and perform SA on those descriptions to determine the nature of the detected activity. By utilizing visual comprehension, context building, and sentiment interpretation, surveillance systems can differentiate between normal and suspicious behavior while reducing false positives and generating actionable insights. Applications include public safety in smart cities, security at high-threat locations such as airports and banks, and monitoring of sensitive areas including hospitals and military installations. This review evaluates how contextual awareness enabled by VLM improves traditional object detection methodologies and supports the transition toward more human-like and explainable alerting modalities. It also discusses limitations related to computational burden, accuracy, and privacy, while highlighting broader societal implications aligned with Sustainable Development Goals focused on urban safety and crime reduction. Future research directions include multimodal fusion, real-time optimization, and ethical guidelines for responsible deployment.

Keywords— Vision-Language Model, Intelligent Surveillance, Anomaly Detection, Object Tracking, Real-Time Alerting.

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	Appendix A: The proposed system “Live Surveillance with Actionable Intelligence” focuses on real-time interpretation of video data using the Salesforce BLIP model and sentiment analysis. To assess the feasibility of this problem, it is important to analyze it using concepts of satisfiability, computational complexity, and mathematical modeling. Appendix B: Details of the papers and Copyright. Appendix C: Plagiarism Report	
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LIST OF ABBREVIATIONS

Abbreviation	Illustration
VLM	Vision Language Model
NLP	Natural Language Processing
LLM	Large Language Model
CCTV	Closed Circuit Television
FPS	Frames per Second
GPU	Graphics Processing Unit
TPU	Tensor Processing Unit
IP	Internet Protocol
CV	Computer Vision
CSV	Comma Separated Values
API	Application Programming Interface

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1. INTRODUCTION

1.1. Overview

The increasing demand for intelligent surveillance systems has driven the adoption of AI-based video analysis technologies. Traditional CCTV monitoring heavily depends on human supervision, making it prone to fatigue, oversight, and delayed response during emergencies. The proposed **Vision-Language Model (VLM)-based Real-Time Surveillance Framework** addresses these challenges by integrating computer vision and natural language understanding to generate descriptive insights from live video feeds. The framework combines frame extraction, image captioning, and sentiment analysis to interpret CCTV footage contextually identifying suspicious, violent, or abnormal activities with minimal human intervention.

1.2. Motivation

- Conventional surveillance systems generate massive amounts of video data that cannot be monitored effectively by humans.
- Most existing automated systems rely only on object detection and lack semantic understanding of complex scenes.
- There is a need for an intelligent, context-aware system that can describe visual scenes in natural language and assess their sentiment to detect unusual or threatening situations.
- Integrating Vision-Language Models (VLMs) with sentiment analysis can bridge the gap between perception and reasoning, making surveillance systems proactive and interpretable.

1.3. Problem Statement

Problem Statement:

Current CCTV surveillance systems are predominantly passive; they record and store video data without providing real-time semantic understanding or incident detection. Existing vision-based models can recognize objects or actions but fail to describe events contextually or infer intent. The objective of this work is to develop a VLM-powered surveillance framework capable of analyzing live or recorded CCTV footage, generating detailed captions for each frame, and evaluating the emotional or situational sentiment of the described scene to identify potential abnormal or harmful events.

Objectives:

- Utilize Vision-Language Models to generate natural-language captions describing each CCTV frame.
- Incorporate a sentiment analysis model to evaluate scene context and categorize the

situation (e.g., normal, suspicious, violent, or distressful).

- Enable real-time video feed processing for continuous, automated monitoring.
- Provide interpretable text outputs that can assist human operators in quick decision-making.

1.4. Project Scope

The scope of this project includes designing and implementing a real-time video analysis system that integrates Vision-Language Models for contextual image understanding and sentiment analysis for intent classification. The framework processes continuous CCTV input, extracts key frames, generates descriptive captions, and analyzes their sentiment to detect potential abnormalities. It is applicable to smart city surveillance, traffic monitoring, public safety, and industrial security systems.

Limitations:

- Real-time processing is dependent on hardware acceleration (GPU/TPU availability).
- Sentiment interpretation is limited by model accuracy and context ambiguity.
- Live deployment requires stable network conditions and efficient video capture pipelines.
- The model currently analyzes frames sequentially and may require optimization for high-FPS streams.

1.5. Methodologies of Problem Solving

The development of the Real-Time Surveillance Framework follows a modular and data-driven methodology that integrates vision-language reasoning with sentiment understanding. The entire process is divided into five systematic stages ensuring structured analysis, implementation, and evaluation.

Step 1 – Requirement Analysis and Problem Understanding

- Study existing surveillance techniques and their limitations in semantic understanding.
- Identify requirements for real-time scene interpretation and incident detection.
- Define expectations such as context awareness, descriptive captioning, and sentiment-based alerting.

Step 2 – Data Collection and Pre-processing

- Capture or import CCTV footage and extract frames at fixed intervals.
- Perform basic preprocessing such as resizing and normalization for model compatibility.
- Store extracted frames in a structured format for further processing.

Step 3 – System Design and Architecture Formulation

- Design a modular architecture comprising three main components:
 - Frame Extractor: Captures and saves frames from live or recorded video streams.
 - Caption Generator (VLM): Generates detailed textual descriptions for each frame using a Vision-Language Model (e.g.,Salesforce's BLIP).
 - Sentiment Analyzer: Classifies the emotional tone or situational sentiment of generated captions.
- Establish data flow between modules to ensure synchronized and continuous video processing

Step 4 – Model Integration and Implementation

- Integrate the Salesforce's BLIP model for caption generation using image-text conditioning.
- Implement a Roberta-based sentiment analysis pipeline to interpret scene context.
- Combine both modules into a single workflow that processes video frames sequentially and generates corresponding descriptive outputs.
- Ensure device optimization for GPU utilization and low-latency inference.

Step 5 – Evaluation and Optimization

- Evaluate the framework on multiple CCTV scenarios (normal, accident, fire, theft, violence, etc.).
- Measure performance in terms of accuracy, latency, and contextual relevance.
- Optimize frame extraction rate and model inference parameters for smooth real-time operation.
- Collect qualitative feedback from human evaluators to assess interpretability and usefulness.

2. LITERATURE SURVEY

Sr No	Paper Title	Journal / Source	Authors & Publication Date	Methodology
1	Vision transformer embedded video anomaly detection using attention driven recurrence	Elsevier	Ummay Maria Muna; 2025	Embeds Vision Transformers with recurrence for contextual video anomaly detection.
2	Salesforce's BLIP: Redefining small and efficient multimodal models	arXiv preprint	Andrés Marafioti, Orr Zohar; 2025	Employs a novel context-aware VLM design for anomaly detection.
3	Joint Feature Extraction and Alignment in Object Tracking with Vision-Language Models	Elsevier	Hong Zhu, Qingyang Lu Lei Xue; 2024	Proposes a joint feature extraction and alignment framework combining visual and language embeddings to improve accuracy and robustness in object tracking.
4	An Introduction to Vision-Language Modeling	arXiv preprint	Florian Bordes, Richard Yuanzhe Pang 2024	Uses a multimodal foundation model for computer vision tasks.
5	ViViT: A Video Vision Transformer	ICCV	Anurag Arnab, Mostafa Dehghani, Georg Heigold; 2024	Proposes a pure Transformer model (ViViT) for video understanding.
6	Joint Visual Grounding and Tracking with Natural Language Specification	CVPR	Zhou et al.; 2024	Combines natural language grounding with tracking to align visual objects and linguistic descriptions.
7	Visual Grounding With Joint Multimodal Representation and Interaction	ICCV	Hong Zhu , Qingyang Lu.; 2023	Introduces a joint multimodal representation (vision + language) with interaction modules for improved visual grounding.

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8	PRAT: Accurate Object Tracking Based on Progressive Attention	arXiv preprint	Yulin Zeng, Bi Zeng; 2023	Proposes a progressive attention mechanism to improve accuracy and robustness in object tracking.
9	Cross-Modal Target Retrieval for Tracking by Natural Language	CVPR	Yihao Li, Jun Yu, Zhongpeng Cai, Yuwen Pan; 2022	Introduces cross-modal retrieval for object tracking guided by language.
10	CoCa: Contrastive Captioners are Image-Text Foundation Models	arXiv preprint	Jiahui Yu, Zirui Wang; 2022	Applies contrastive vision-language pretraining for downstream tasks.
11	Towards More Flexible and Accurate Object Tracking with Natural Language: Algorithms and Benchmark	CVPR	Xiao Wang ¹ , Xiujun Shu ² ; 2021	Provides algorithms and a benchmark dataset for natural language-guided object tracking.
12	TrTr: Visual Tracking with Transformer	arXiv preprint	Moju Zhao, Kei Okada; 2021	Introduces a transformer-based approach (details missing)
13	ViLT: Vision-and-Language Transformer Without Convolution or Region Supervision	CVPR	Wonjae Kim, Bokyung Son, Ildoo Kim; 2021	Simplifies VLM by directly embedding image patches and text tokens into a transformer.

3. SOFTWARE REQUIREMENT SPECIFICATION

3.1. Assumptions and Dependencies

- Continuous video input available from CCTV or live IP camera feed.
- Internet connectivity for downloading pretrained Vision-Language and Sentiment models (initial setup).
- GPU support (CUDA) or high-performance CPU for real-time frame analysis and model inference.
- Access to compatible Python libraries such as Transformers, Torch, OpenCV, and Rich for processing and visualization.
- Storage availability for temporary frame extraction and log files.

3.2. Functional Dependencies

- **Frame Extraction:** Continuously capture frames from a live CCTV feed or recorded video at defined intervals for analysis.
- **Caption Generation (VLM):** Use a Vision-Language Model (e.g., Salesforce's BLIP) to generate descriptive captions that summarize each frame's context and activities.
- **Sentiment Analysis:** Process generated captions using a Roberta-based sentiment classifier to determine the emotional or situational tone (e.g., positive, neutral, negative).
- **Event Detection and Alerting:** Trigger audible or visual alerts when the sentiment analysis detects potential negative or alarming activities (e.g., accidents, violence, or fire).
- **Real-Time Monitoring Pipeline:** Enable continuous frame extraction, caption generation, and sentiment evaluation in an automated loop for live surveillance feeds.
- **Visualization and Reporting:** Display processed results (frame name, caption, sentiment, confidence score) in the terminal with color-coded outputs using the Rich library.
- **Data Management:** Automatically delete processed frames to optimize storage and maintain continuous real-time operation.
- **Logging and Traceability:** Store key event logs (timestamp, sentiment status, description) for later analysis and verification.
- **Modularity and Extensibility:** Each component (frame extractor, caption generator, sentiment analyzer) is independent, allowing future upgrades or model replacements without system overhaul.

3.3.External Interface Requirements

User Interface:

- Command-line or console-based interface displaying live captions and sentiment insights.
- Optional future integration with a dashboard for visual monitoring.

Hardware Interfaces:

- CCTV or IP camera for live video feed input.
- Standard computing hardware (PC/server) with webcam compatibility.

Software Interfaces:

- Vision-Language Model API (e.g., HuggingFace Salesforce's BLIP).
- Transformer-based sentiment analysis pipeline (e.g., Roberta).
- OpenCV for video frame capture and management.

Communication Interfaces:

- Local data flow between modules via Python function calls and file I/O.
- Option for future RESTful API integration for cloud-based monitoring

3.4.Non-Functional Requirements

Performance:

- Average frame processing time < 3 seconds for smooth real-time feedback.
- Low latency caption generation and sentiment classification.

Scalability:

- Systems can handle multiple camera feeds with minor configuration adjustments.
- Modular structure supports distributed or cloud-based deployment in future versions.

Security:

- Restricted access to live video feeds and processed data.
- No external sharing of surveillance footage unless authorized.

Reliability:

- Continuous operation with automatic restart in case of frame or model failure.
- Temporary caching to handle short network or camera disconnections.

Maintainability:

- Code structured into modular Python files for easy debugging and upgrades.
- Clear separation of model, logic, and visualization components.

Auditability:

- Each event logged with timestamp, caption, and sentiment label for post-analysis verification.

3.5. System Requirements

Database:

- Local log storage for event records.
- Temporary folder structure for extracted frames.

Software Stack:

- Backend: Python 3.10+, OpenCV for video input, Transformers for model inference, Torch for deep learning operations.
- Models: Salesforce's BLIP for vision-language captioning, Roberta for sentiment analysis.
- Visualization: Rich (for terminal output).

Hardware:

- Development Environment: CPU with 8 cores, 16 GB RAM.
- Optimal Runtime: NVIDIA GPU (RTX 3060 or above) for faster model inference.
- Camera Input: Any standard webcam or CCTV IP feed

3.6. Analysis Models: SDLC Model to be applied

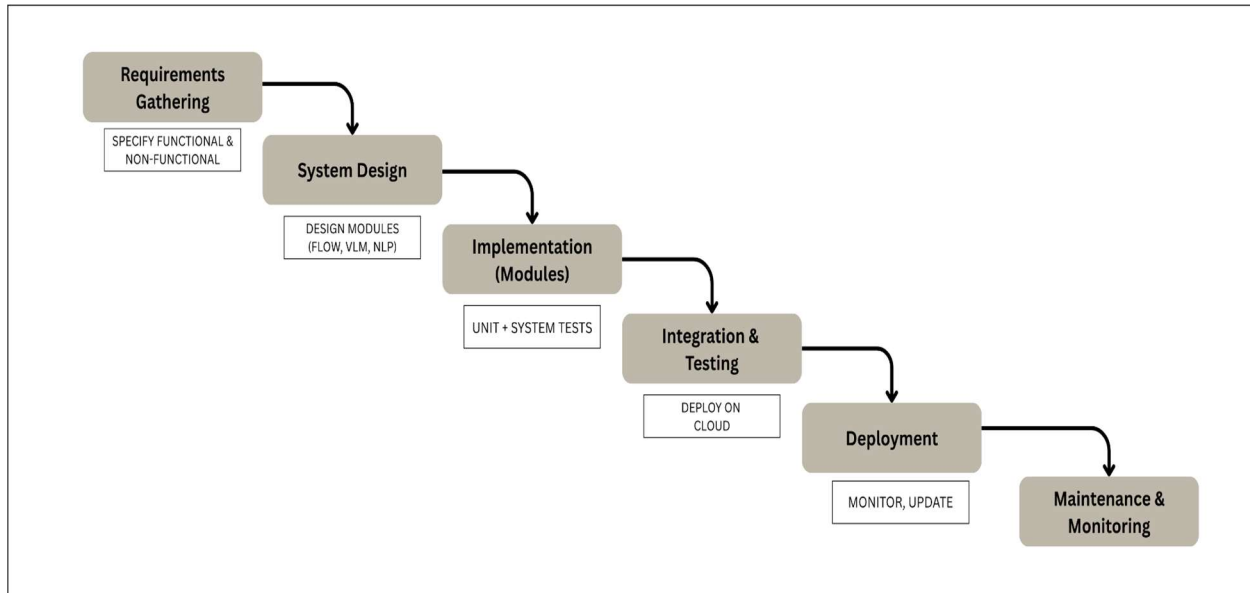


Figure 3.6.1: Waterfall Diagram

3.7. System Implementation Plan

1. Requirement Gathering and Analysis:

- Identify the core users – security personnel, surveillance operators, and smart city monitoring units.
- Define both functional requirements (real-time frame extraction, caption generation, sentiment detection, alerts) and non-functional requirements (performance, scalability, reliability).
- Determine hardware (camera, GPU-enabled systems) and software dependencies (Torch, Transformers, OpenCV, Rich).
- Establish data handling requirements for temporary storage, logs, and event alerts.

2. System Design:

- Design a modular architecture consisting of four main components:
- Frame Extractor Module – Handles real-time frame acquisition from video feed.
- Caption Generator (VLM) – Generates descriptive text for each frame.
- Sentiment Analyzer – Determines emotional tone or urgency level.
- Alert Manager & Console Visualizer – Displays results and triggers alerts.
- Develop DFD Level 0–2 diagrams to represent data flow between modules.
- Create UML diagrams such as class and sequence diagrams to illustrate interaction between model layers.
- Define data structures for storing temporary frames, generated captions, and sentiment results.

3. Implementation:

- Phase 1: Develop the frame extraction pipeline using OpenCV to read frames at fixed intervals from a video or live camera.
- Phase 2: Integrate Salesforce's BLIP for caption generation from each extracted frame.
- Phase 3: Implement the Roberta-based sentiment analyzer to evaluate each caption's tone.
- Phase 4: Build the alert system to trigger sounds and color-coded messages for negative or critical sentiments.
- Phase 5: Optimize code for batch frame handling and continuous feed processing.
- Phase 6: Enhance logging functionality to store event timestamps and detected anomalies.
- Phase 7: (Future Scope) Integrate a live dashboard or web interface for visual surveillance management.

4. Testing:

- **Unit** **Testing:**
Validate each module individually — frame extraction, caption generation, sentiment analysis, and alert triggers.
- **Integration** **Testing:**
Ensure smooth data flow between frame extraction, captioning, and sentiment modules for continuous real-time processing.
- **Performance** **Testing:**
Measure average frame processing speed, caption latency, and sentiment accuracy.
- **Stress** **Testing:**
Simulate multiple video streams or prolonged runtime to ensure system stability.
- **User Acceptance Testing (UAT):**
Conduct field testing with CCTV operators or demo users to validate usability and alert reliability.

5. Deployment:

- Deploy the system on a **local workstation** or **cloud-based GPU server**.
- Configure the pipeline for live video input (e.g., RTSP/IP camera).
- Ensure dependency management via virtual environments and model caching for faster startup.
- Enable logging and alerting modules to operate automatically during runtime.

6. Maintenance:

- Monitor system logs and performance metrics to detect anomalies or hardware failures.
- Periodically update the Vision-Language Model and sentiment classifier for improved accuracy.
- Release patches for bug fixes, frame rate optimization, and new camera compatibility.
- Extend functionality by integrating external APIs (e.g., emergency response notification or cloud dashboards).
- Maintain documentation for future scalability and model fine-tuning efforts.

4. SOFTWARE REQUIREMENT SPECIFICATION

4.1. System Architecture

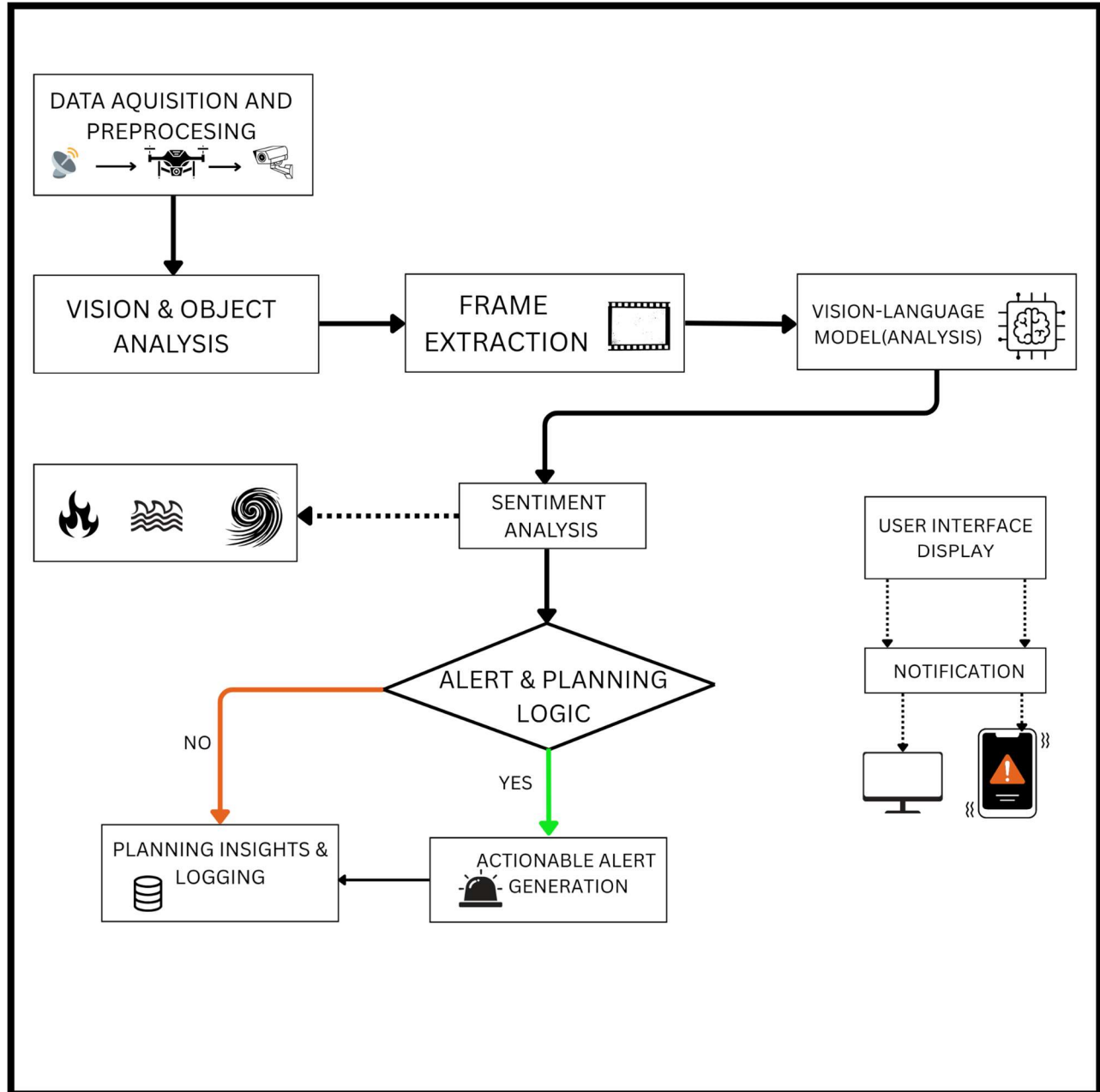


Figure 4.1: System Architecture

Components:

- **Live Stream / Video Input Module:** Captures real-time video feed or accepts pre-recorded video input for analysis.
- **Frame Extraction Module:** Processes video stream into frames at defined intervals for efficient analysis.
- **Visual-Language Model (VLM) Caption Generator:** Generates descriptive captions or scene context from each extracted frame using the integrated multimodal model.
- **Sentiment Analysis Module:** Analyzes generated captions to classify the emotional tone (positive, negative, or neutral) using a fine-tuned sentiment model.
- **Context Fusion Layer:** Combines multiple frame-level sentiments and captions to form an overall contextual understanding of the live stream.
- **Real-Time Output Visualizer:** Displays live updates of captions, sentiment results, and trends in an interactive dashboard or terminal feed.
- **Data Storage & Logging Module:** Stores processed frames, captions, and sentiment data for further review or retraining.
- **System Control & Monitoring Layer:** Manages synchronization between modules, ensures smooth frame processing, and handles errors or delays in real-time.

4.2. Data Flow Diagrams

DFD level 0:

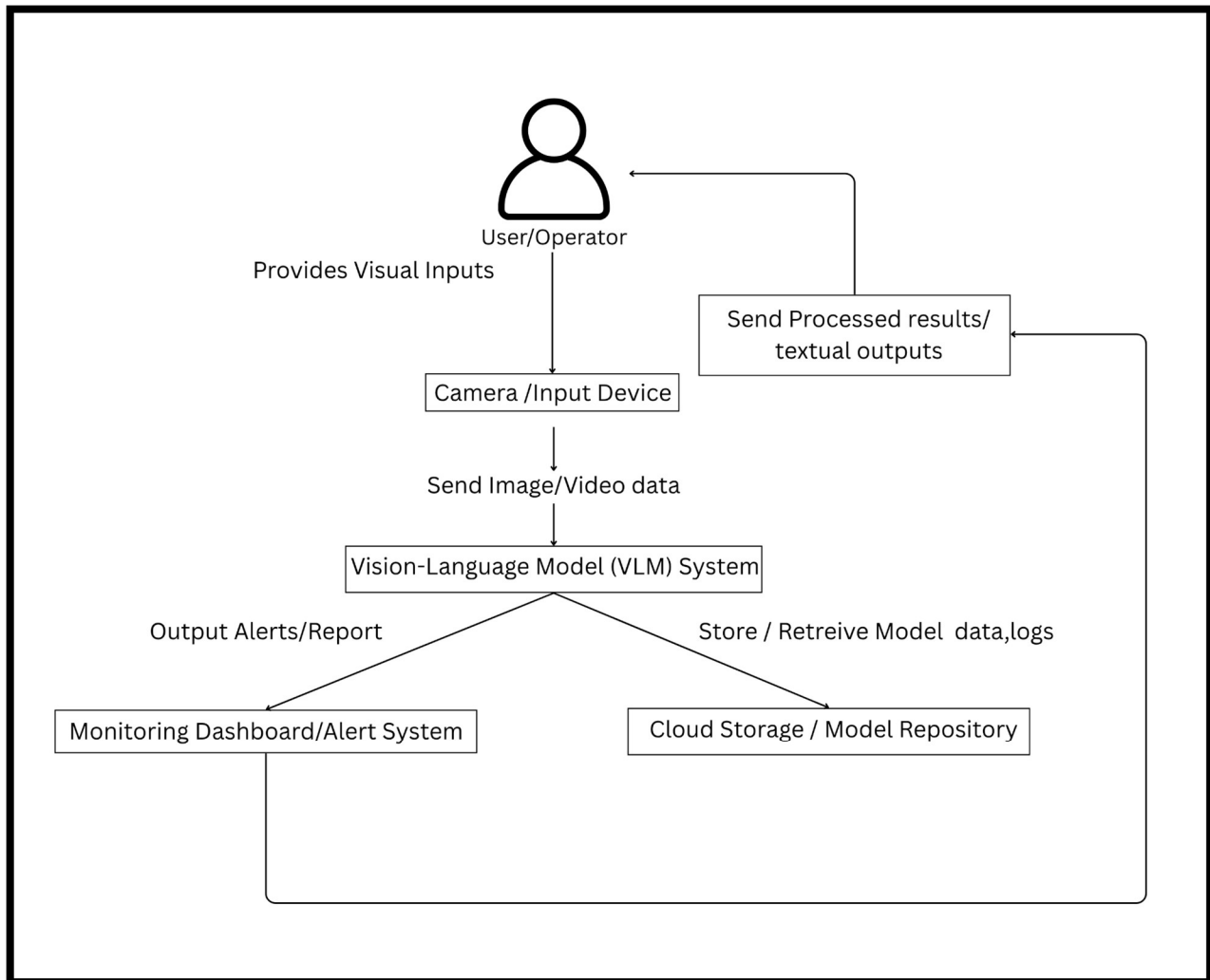


Figure 4.2.1: DFD Level 0

DFD Level 1

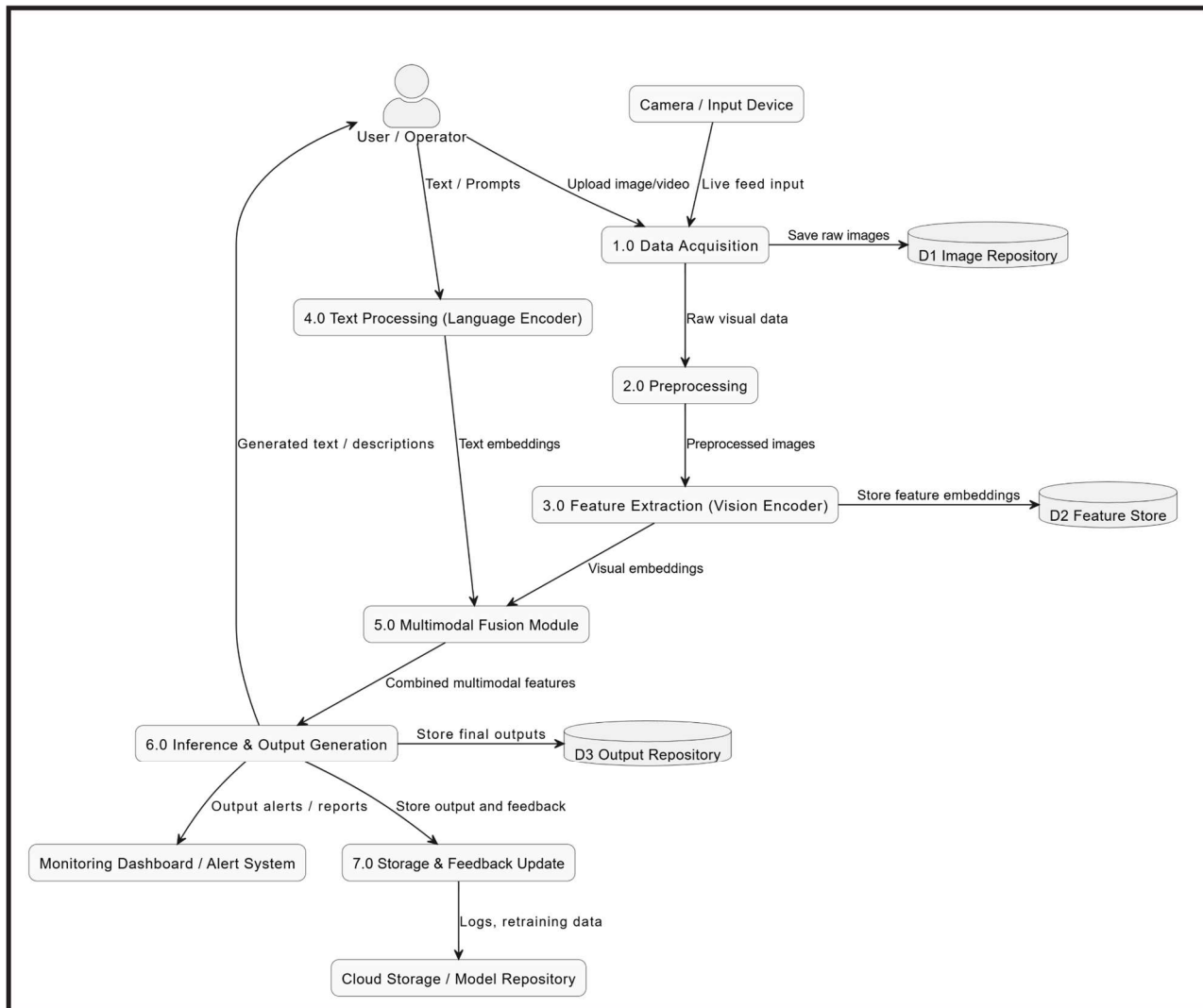


Figure 4.2.2: DFD Level 1

DFD level 2

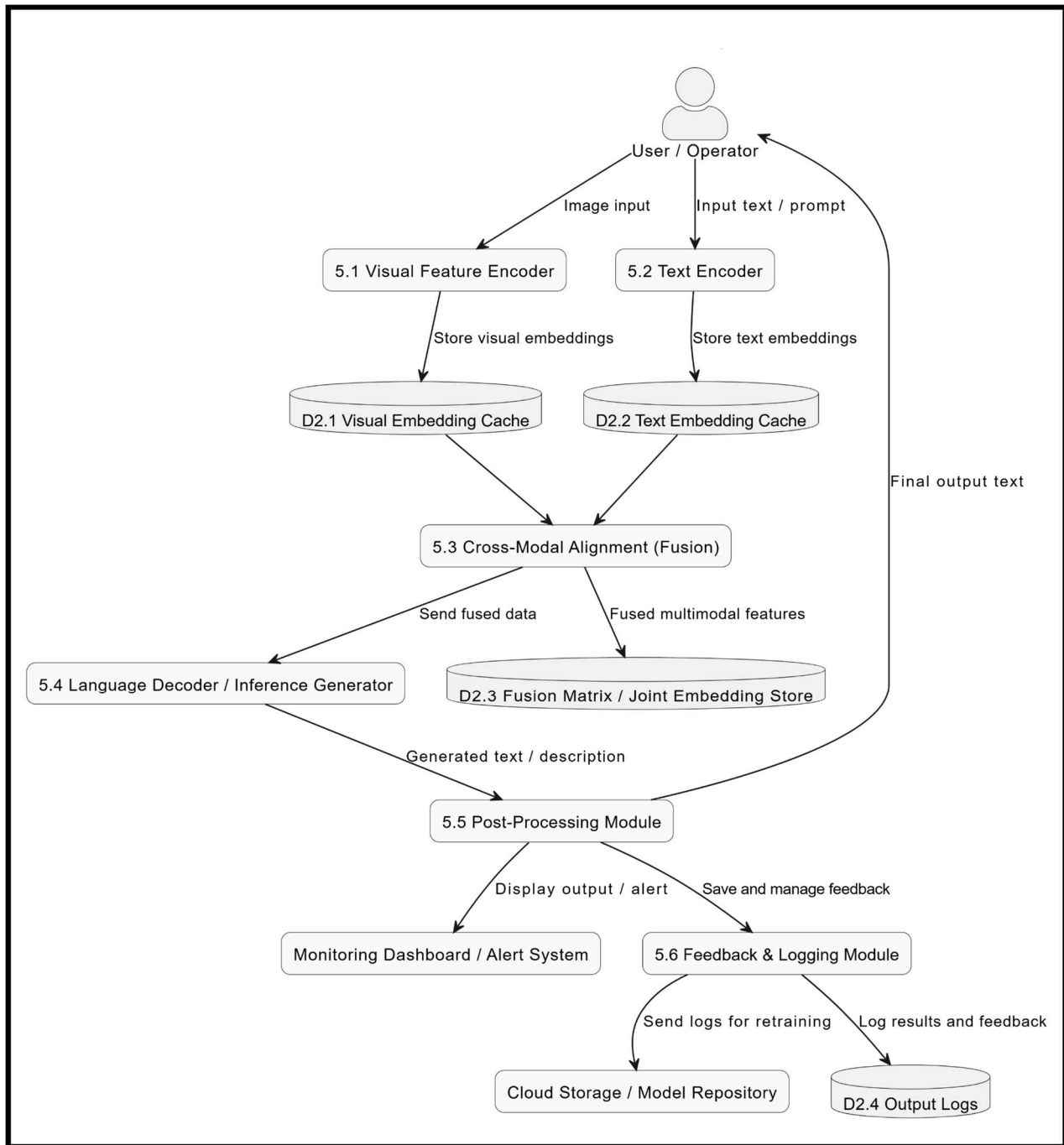


Figure 4.2.3: DFD Level 2

4.3. ER Diagram

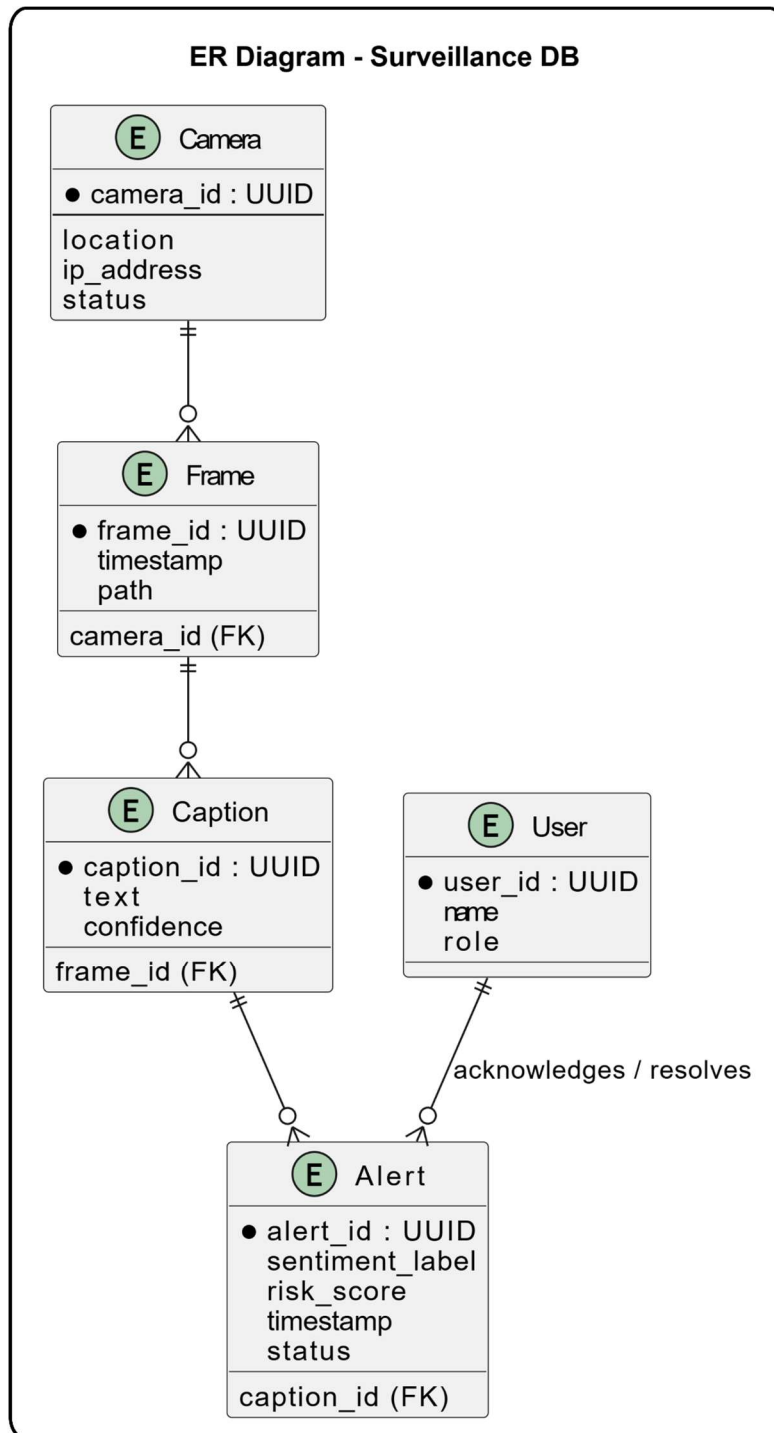


Figure 4.3: ER Diagram

4.1 UML-DIAGRAM

4.1.1 USE CASE DIAGRAM

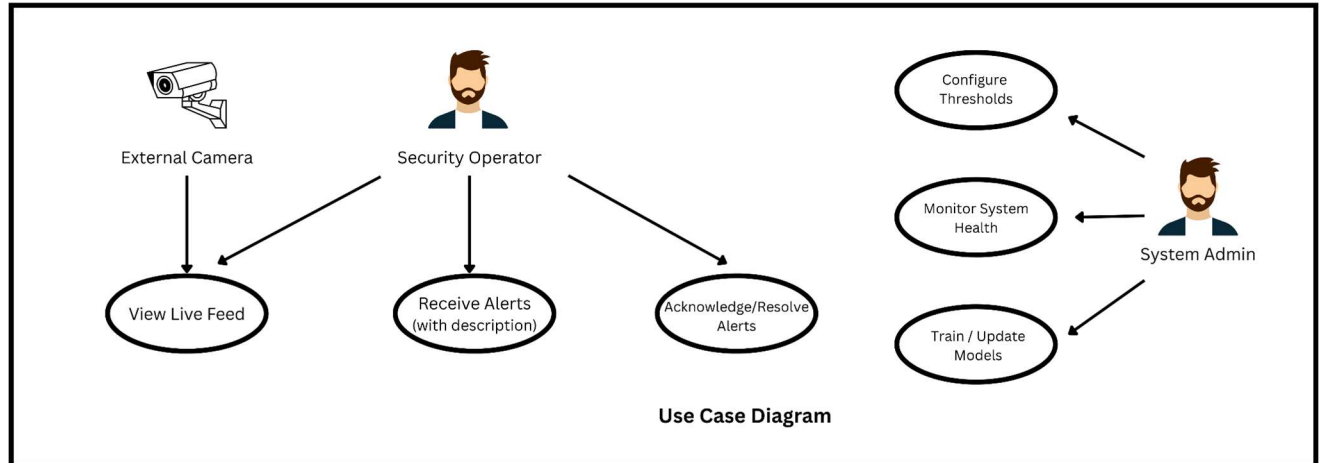


Figure 4.4.1: Sequence Diagram

4.1.2 SEQUENCE DIAGRAM

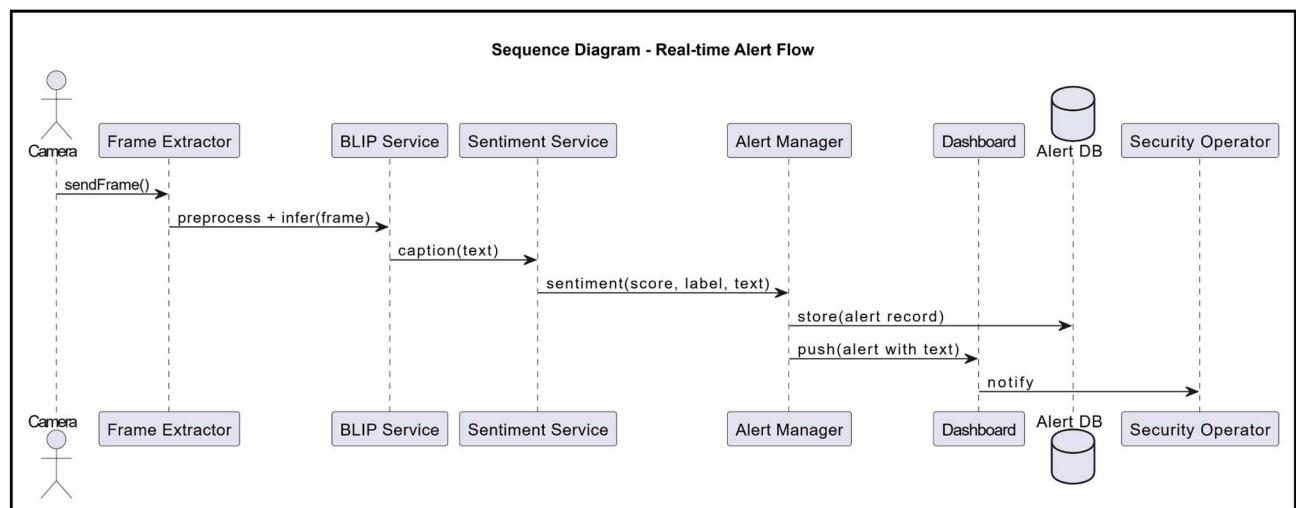


Figure 4.4.2: Sequence Diagram

4.1.3 COLLABORATIVE DIAGRAM

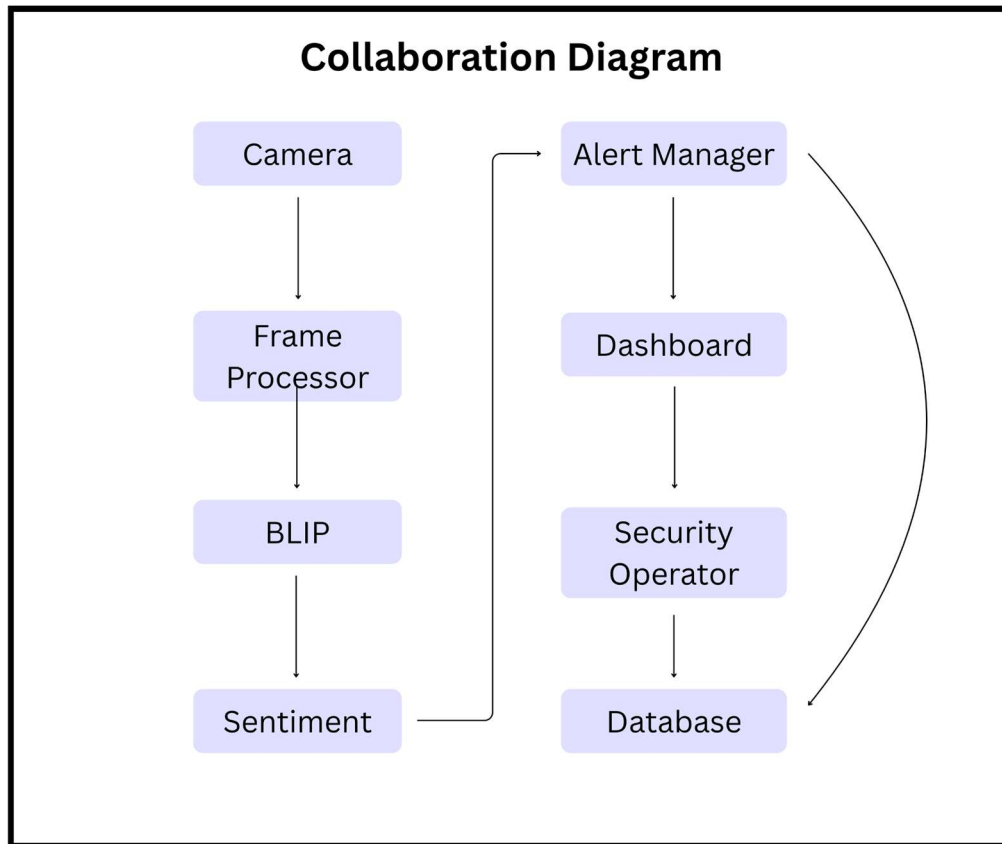


Figure 4.4.3: Collaborative Diagram

4.1.4 DEPLOYMENT DIAGRAM

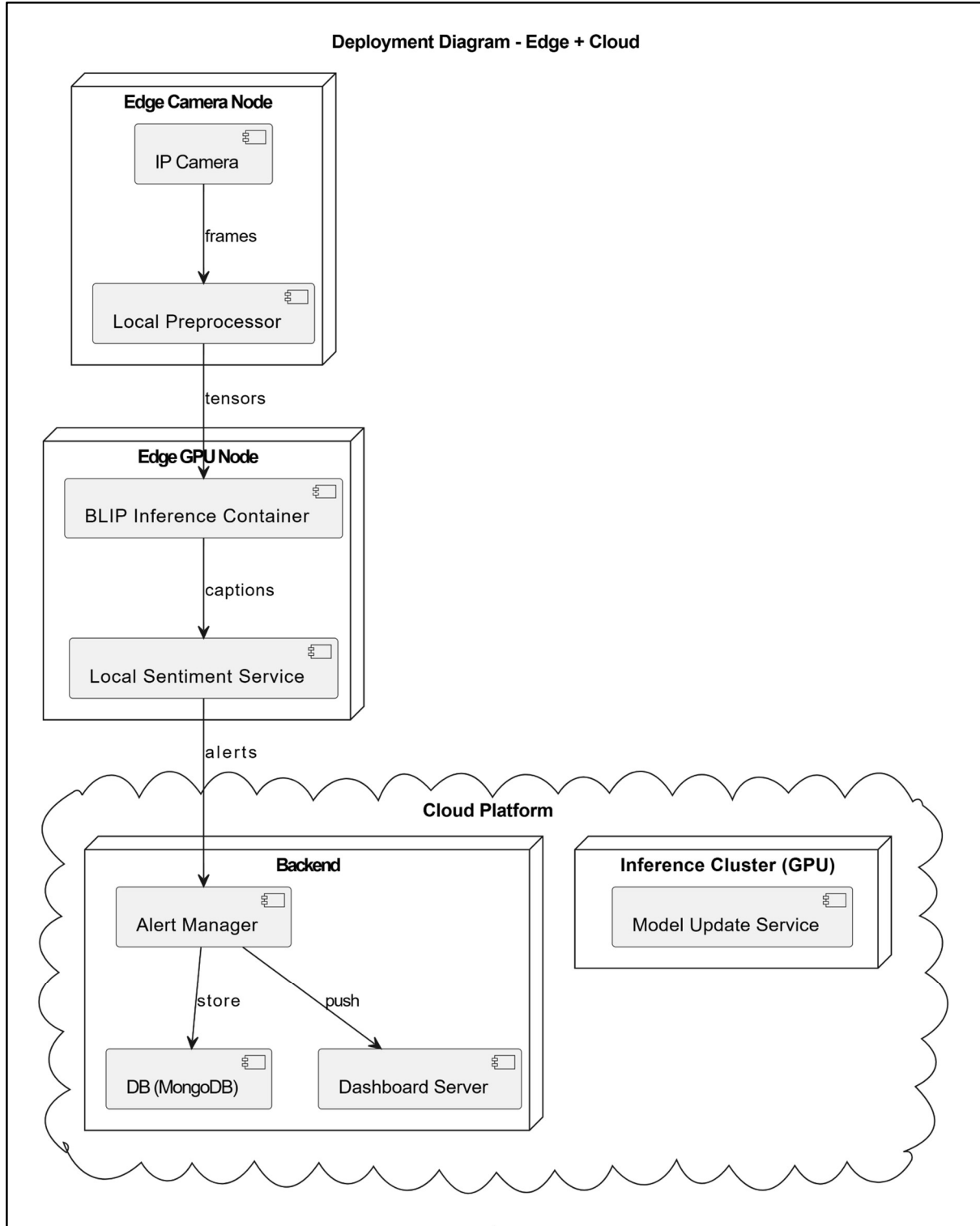


Figure 4.4.4: Deployment Diagram

4.1.5 COMPONENT DIAGRAM

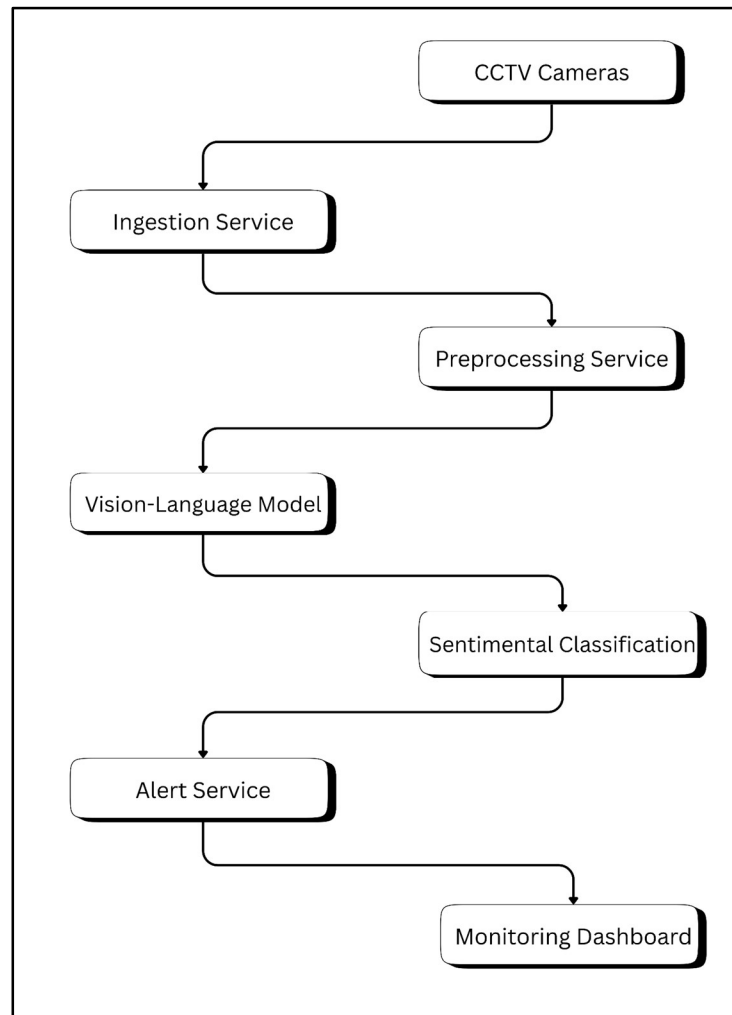


Figure 4.4.5: Component Diagram

4.1.6 CLASS DIAGRAM

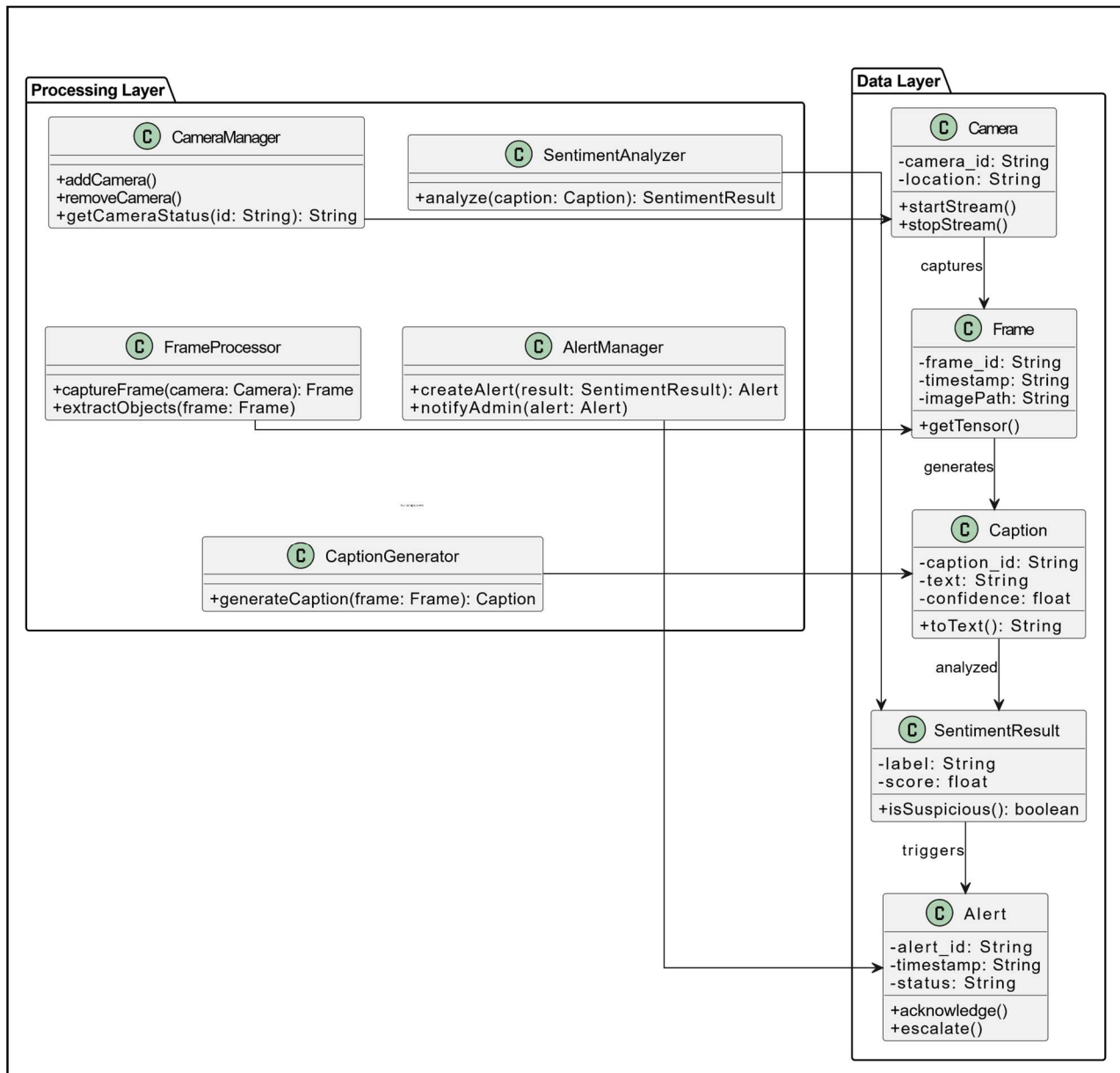


Figure 4.4.6: Class Diagram

4.1.7 STATE CHART DIAGRAM

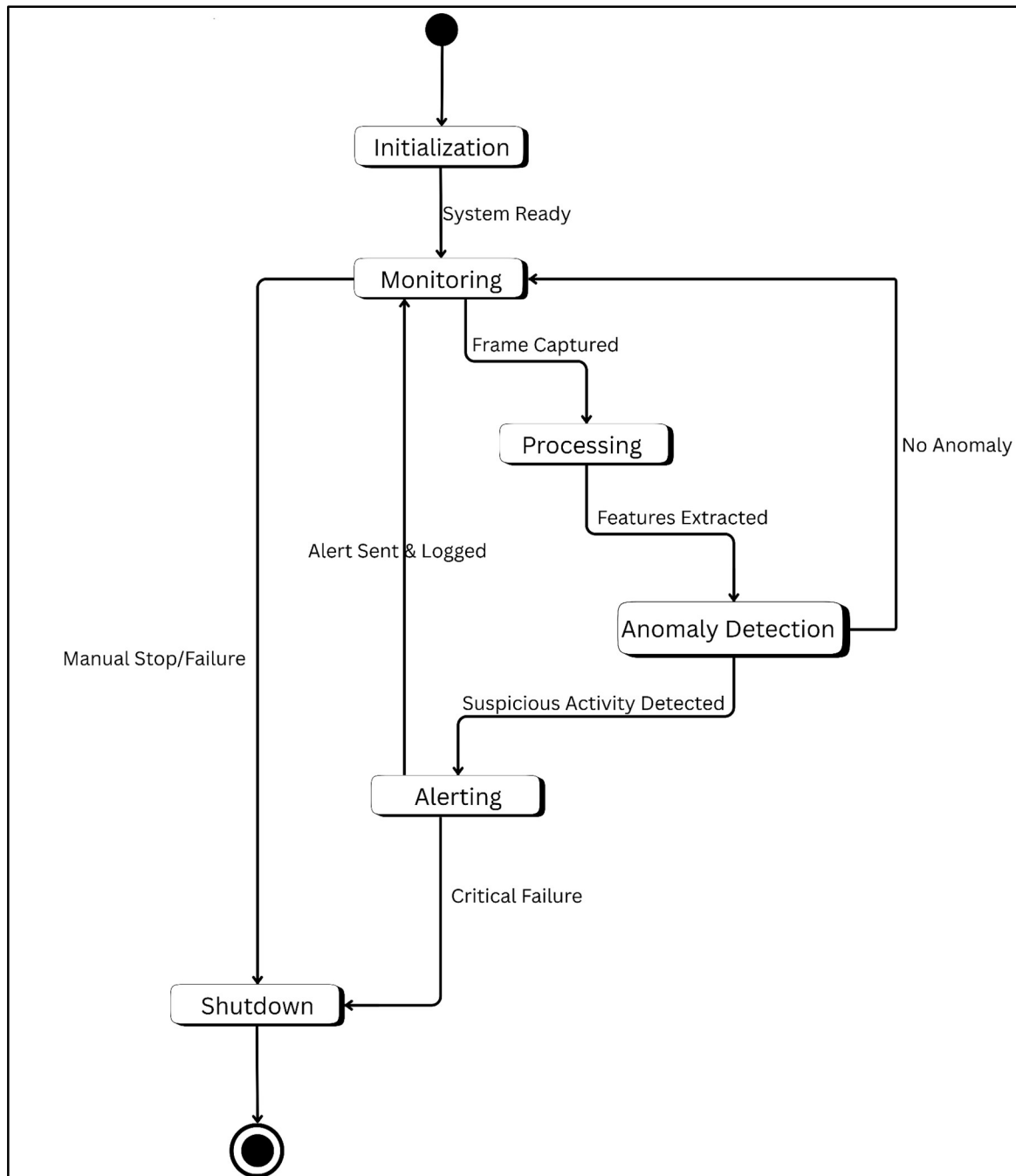


Figure 4.4.7: State Chart Diagram

4.1.8 ACTIVITY DIAGRAM

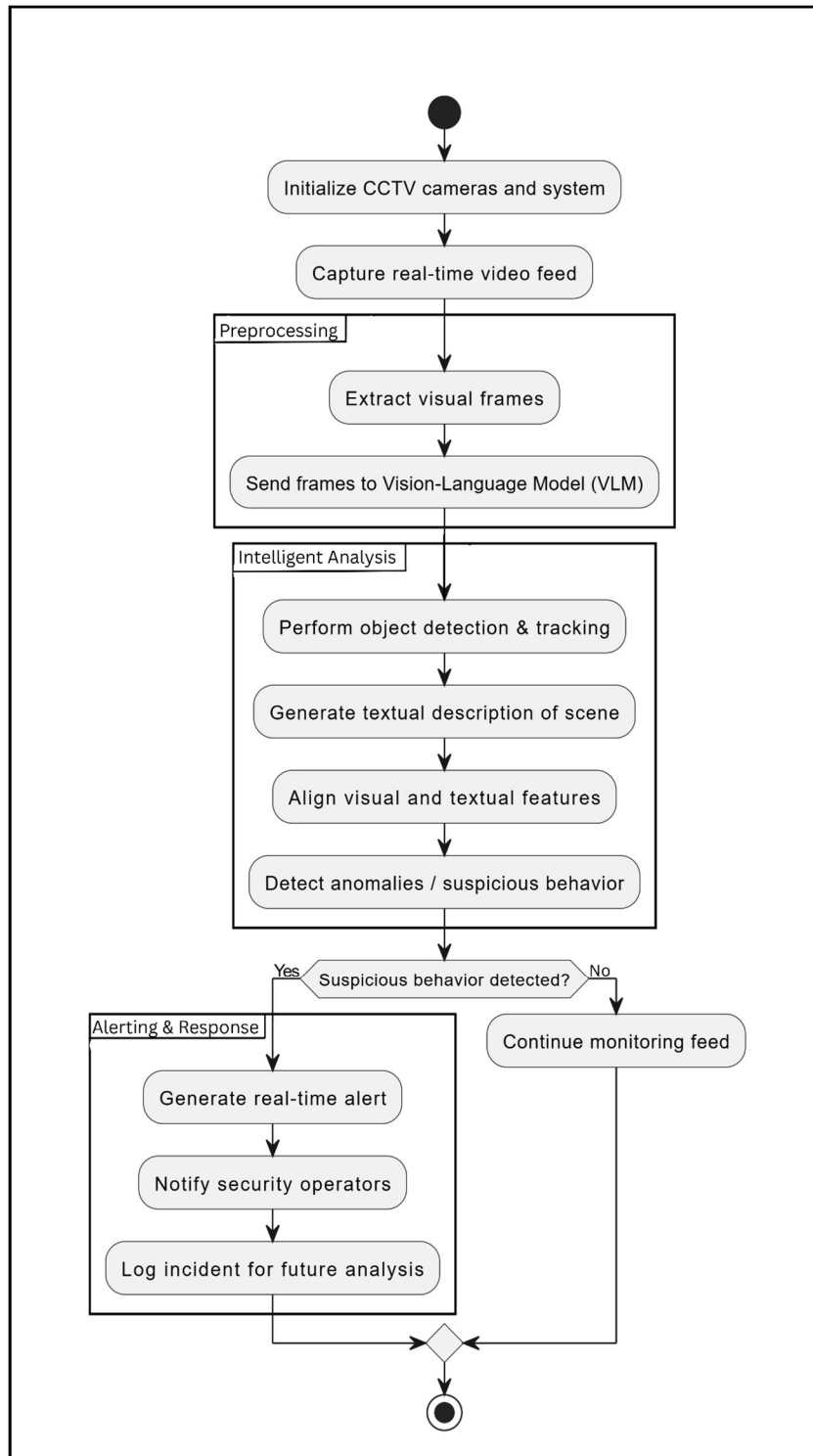


Figure 4.4.8: Activity Diagram

5. PROJECT PLAN

5.1. Project Estimate

The project “Live Surveillance with Actionable Intelligence” is designed as an AI-based surveillance system that leverages the Salesforce BLIP model for vision-language understanding. The system transforms real-time video input into meaningful textual descriptions, enabling contextual interpretation of activities.

The project estimation mainly involves computational resources, development effort, and implementation time. Since the BLIP model performs image captioning using deep learning, the system requires moderate to high computational capability, especially for real-time inference.

The development process includes video acquisition, preprocessing, BLIP-based caption generation, sentiment analysis, and alert generation. By using pre-trained BLIP models and open-source frameworks, the project minimizes cost while maintaining high performance and accuracy.

5.1.1. Reconciled Estimates

The reconciled estimate focuses on optimizing cost while effectively utilizing the capabilities of the Salesforce BLIP model. Since BLIP is available as a pre-trained model, the need for extensive training is eliminated, reducing both time and computational expenses.

The key considerations include:

- **Hardware** **Cost:**
A GPU-enabled system is recommended for efficient BLIP inference and real-time processing.
- **Software** **Cost:**
Minimal, as BLIP and supporting tools like PyTorch and Hugging Face Transformers are open-source.
- **Development** **Effort:**
Time required for integrating BLIP with video processing and sentiment analysis modules.
- **Performance** **Cost** **Balance:**
BLIP provides high-quality contextual captions without requiring extremely large-scale infrastructure, making it a cost-effective solution.

5.1.2. Project Resources

The implementation of the system using the Salesforce BLIP model requires the following resources:

1. Hardware Resources:

- GPU-enabled system (preferred for BLIP model inference)
- Standard computer for development
- CCTV cameras or video datasets for testing

2. Software Resources:

- Programming Language: Python (3.10+)
- Libraries: OpenCV, PyTorch
- Frameworks: Hugging Face Transformers (for BLIP integration)
- Models:
 - BLIP (for image captioning and vision-language understanding)
 - RoBERTa (for sentiment analysis)
- Development Tools: VS Code or similar IDE
- Storage: CSV/SQLite for logging

3. Human Resources:

- Developers for system design and implementation
- Team members for testing and documentation
- Project guide for supervision

These resources enable the system to effectively utilize the BLIP model for generating semantic captions and performing context-aware surveillance.

5.2. Risk Management

Risk management is a crucial component in the development of the “Live Surveillance with Actionable Intelligence” system. Since the project relies on advanced AI techniques such as the Salesforce BLIP model and transformer-based sentiment analysis, various technical, operational, and ethical risks must be considered.

The system operates in real-time environments, where incorrect predictions, delays, or system failures may impact decision-making. Additionally, the use of AI in surveillance introduces concerns related to privacy, data handling, and model reliability. Therefore, identifying, analyzing, and managing risks is essential to ensure system robustness, accuracy, and responsible deployment.

5.2.1.Risk Identification

The following key risks have been identified in the proposed system:

Model Accuracy Risk: The Salesforce BLIP model may generate incorrect or ambiguous captions, leading to misinterpretation of activities.

Sentiment Misclassification: The sentiment analysis model may incorrectly classify normal activities as suspicious or hostile, or vice versa.

Computational Overhead: High processing requirements of BLIP and transformer models may lead to delays in real-time performance.

Data Quality Issues: Poor video quality, low resolution, or environmental factors (lighting, occlusion) can affect system accuracy.

False Positives/Negatives: Incorrect alert generation may either cause unnecessary panic or miss critical threats.

Privacy and Ethical Concerns: Continuous surveillance and data processing may raise privacy and ethical issues.

System Integration Risk: Difficulty in integrating multiple modules (video processing, BLIP, sentiment analysis, alert system) may affect system stability.

5.2.2.Risk Analysis

Each identified risk is analyzed based on its impact and likelihood:

- **High Impact Risks:**

- Incorrect caption generation and sentiment misclassification can directly affect decision-making.
- Privacy concerns may limit real-world deployment.

- **Moderate Impact Risks:**

- Computational delays may reduce real-time efficiency but can be optimized with hardware support.
- Integration challenges may delay development but are manageable.

- **Low Impact Risks:**

- Minor variations in video quality may slightly affect performance but not overall functionality.

The most critical risks are those affecting accuracy and real-time response, as they directly influence the effectiveness of the surveillance system.

5.2.3. Overview of Risk Mitigation, Monitoring, and Management

To ensure reliable system performance, several strategies are implemented:

- **Mitigation Strategies:**

- Use of pre-trained and well-optimized BLIP models to improve caption accuracy
- Fine-tuning sentiment analysis models for better classification
- Implementing preprocessing techniques (noise reduction, normalization) to improve input quality
- Using GPU acceleration to reduce latency
- Applying threshold-based alert systems to minimize false alarms

- **Monitoring Mechanisms:**

- Continuous logging of system outputs (captions, sentiment, alerts)
- Performance tracking using metrics such as accuracy and latency
- Periodic evaluation using test datasets

- **Management Approaches:**

- Modular system design for easier debugging and updates
- Regular system updates and model improvements
- Incorporation of ethical guidelines and data privacy practices
- Human-in-the-loop validation for critical alerts

Overall, effective risk management ensures that the system remains accurate, efficient, and reliable while addressing practical deployment challenges.

5.3. Project Schedule

5.3.1. Project Task

The project is divided into multiple phases, each consisting of defined tasks executed sequentially to ensure systematic development of the “Live Surveillance with Actionable Intelligence” system.

Phase 1 — Requirement Analysis & Planning

- Identify functional and non-functional requirements of the surveillance system
- Study Vision-Language Models and sentiment analysis techniques
- Analyze existing CCTV surveillance limitations and define project objectives
- Prepare system workflow and implementation strategy

Phase 2 — Data Collection & Preprocessing

- Collect CCTV footage and surveillance video datasets
- Extract frames from video streams at fixed intervals
- Perform preprocessing such as resizing, normalization, and frame organization
- Store processed frames for model inference

Phase 3 — Vision-Language Model Integration

- Integrate Salesforce BLIP model for image caption generation
- Generate contextual descriptions from extracted frames
- Optimize caption generation for real-time inference
- Validate generated captions for contextual relevance

Phase 4 — Sentiment Analysis & Alert System

- Implement Roberta-based sentiment analysis pipeline
- Classify generated captions into positive, neutral, or negative sentiment
- Detect suspicious or abnormal activities using sentiment interpretation
- Develop alert generation mechanism for critical situations

Phase 5 — System Integration & Interface Development

- Integrate frame extraction, BLIP model, and sentiment analysis modules
- Develop console/dashboard interface for displaying outputs
- Implement logging and event management system
- Enable real-time monitoring and continuous processing

Phase 6 — Testing, Optimization & Deployment

- Perform unit and integration testing of all modules
- Evaluate processing speed, latency, and detection accuracy
- Optimize system performance for real-time surveillance

- Deploy the system and finalize project documentation

5.3.2.Task Network

The task network illustrates the sequence and interdependence of activities involved in developing the “Live Surveillance with Actionable Intelligence” system using Vision-Language Models and sentiment analysis. Each phase depends on the successful completion of the previous stage to ensure systematic implementation and integration.

Requirement Analysis → System Design → Data Collection & Preprocessing → BLIP Model Integration → Sentiment Analysis & Alert Generation → System Integration → Testing → Deployment.

Requirement Analysis is the initial phase where project objectives, surveillance requirements, and system functionalities are identified. This is followed by System Design, which defines the architecture, workflow, and interaction between modules such as frame extraction, caption generation, sentiment analysis, and alert handling.

The Data Collection & Preprocessing phase involves gathering CCTV footage, extracting frames, and preparing image data for processing. The Development phase includes integrating the Salesforce BLIP model for caption generation, implementing the Roberta sentiment analysis model, and developing the alert mechanism for abnormal event detection.

The System Integration phase combines all modules into a continuous real-time surveillance pipeline with monitoring and logging features. Finally, Testing and Deployment are performed to validate system performance, optimize accuracy, and ensure reliable real-time operation.

This step-by-step workflow ensures proper planning, systematic development, efficient monitoring, and timely completion of the project.

5.3.3. Timeline Chart

							PHASE ONE			PHASE TWO		
1	Project Research & Planning	AS	DAY 1	DAY 21	21	90						
1.1	Literature Review	RK	DAY 1	DAY 7	7	90%						
1.2	Problem Definition	AS	DAY 5	DAY 10	6	80%						
1.3	Dataset Identification	TS	DAY 8	DAY 14	7	70%						
1.4	System Architecture Planning	TS	DAY 12	DAY 21	10	60%						
2	Data Collection & Preparation	AS	DAY 22	DAY 42	20	50%						
2.1	Dataset Collection	AV	DAY 22	DAY 30	8	40%						
2.2	Frame Extraction from Videos	TS	DAY 26	DAY 36	10	30%						
2.3	Data Cleaning	RK	DAY 30	DAY 35	5	70%						
2.4	Dataset Formatting	AV	DAY 34	DAY 40	6	50%						
3	Model Development & Integration	TS	DAY 43	DAY 70	27	50%						
3.1	Image Caption Model Implementation	AS	DAY 43	DAY 50	7	60%						
3.2	Sentiment Analysis Model	TS	DAY 50	DAY 57	7	70%						
3.3	Object Detection Integration	AS	DAY 55	DAY 60	5	80%						
3.4	System Pipeline Integration	AS	DAY 63	DAY 70	7	80%						
4	Testing, Evaluation & Documentation	AV	DAY 71	DAY 86	15	40%						
4.1	System Testing	TS	DAY 75	DAY 90	15	70%						
4.2	Performance Evaluation	AV	DAY 80	DAY 95	15	80%						
4.3	Report Writing	RK	DAY 85	DAY 100	15	90%						
4.4	Project Updates	AV	DAY 90	DAY 105	15	75%						
4.5	Chart Updates	AS	DAY 96	DAY 105	9	90%						
5	Project Performance / Monitoring	RK	DAY 100	DAY 110	10	80%						
5.1	Project Objectives	AV	DAY 110	DAY 120	10	85%						
5.2	PPT & Final Submission	ALL	DAY 115	DAY 130	15	100%						

							PHASE THREE			PHASE FOUR		
1	Project Research & Planning	AS	DAY 1	DAY 21	21	90						
1.1	Literature Review	RK	DAY 1	DAY 7	7	90%						
1.2	Problem Definition	AS	DAY 5	DAY 10	6	80%						
1.3	Dataset Identification	TS	DAY 8	DAY 14	7	70%						
1.4	System Architecture Planning	TS	DAY 12	DAY 21	10	60%						
2	Data Collection & Preparation	AS	DAY 22	DAY 42	20	50%						
2.1	Dataset Collection	AV	DAY 22	DAY 30	8	40%						
2.2	Frame Extraction from Videos	TS	DAY 26	DAY 36	10	30%						
2.3	Data Cleaning	RK	DAY 30	DAY 35	5	70%						
2.4	Dataset Formatting	AV	DAY 34	DAY 40	6	50%						
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3.2	Sentiment Analysis Model	TS	DAY 50	DAY 57	7	70%						
3.3	Object Detection Integration	AS	DAY 55	DAY 60	5	80%						
3.4	System Pipeline Integration	AS	DAY 63	DAY 70	7	80%						
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4.1	System Testing	TS	DAY 75	DAY 90	15	70%						
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4.3	Report Writing	RK	DAY 85	DAY 100	15	90%						
4.4	Project Updates	AV	DAY 90	DAY 105	15	75%						
4.5	Chart Updates	AS	DAY 96	DAY 105	9	90%						
5	Project Performance / Monitoring	RK	DAY 100	DAY 110	10	80%						
5.1	Project Objectives	AV	DAY 110	DAY 120	10	85%						
5.2	PPT & Final Submission	ALL	DAY 115	DAY 130	15	100%						

5.4.Team Organization

The project was developed collaboratively by a team where responsibilities were distributed among members.

5.4.1.Team structure

The team structure consisted of members assigned to specific tasks:

- Dataset Management: Dataset preparation and annotation
- Model Development: GAN training and segmentation model development
- System Integration: Developing the Streamlit interface and integrating the model
- Evaluation and Documentation: Performance evaluation and report preparation Phase.

5.4.2.Management reporting and communication

Regular communication and reporting were maintained throughout the project. The team followed these communication methods:

- Weekly team meetings to discuss progress
- Task allocation among team members
- Regular updates to the project guide or supervisor
- Documentation of development milestones This ensured smooth coordination and successful completion of the project.

6. PROJECT IMPLEMENTATION

The implementation of the “Live Surveillance with Actionable Intelligence” system is based on a modular pipeline that integrates computer vision, vision-language modeling, and sentiment analysis. The system processes real-time video streams, converts visual data into descriptive text using the Salesforce BLIP model, and evaluates contextual risk through sentiment analysis.

Each module is designed independently and then integrated to ensure scalability, maintainability, and efficient real-time performance.

6.1. Overview of Project Modules

The system is divided into the following key modules:

- **Video Acquisition Module:** Captures live video streams from CCTV cameras or datasets.
- **Preprocessing Module:** Extracts frames from video and applies noise reduction, normalization, and resizing to improve input quality.
- **Vision-Language Module (BLIP):** Uses the Salesforce BLIP model to generate descriptive captions from each frame, enabling semantic understanding of activities.
- **Sentiment Analysis Module:** Processes generated captions using a transformer-based model (e.g., RoBERTa) to classify behavior as Normal, Suspicious, or Hostile.
- **Alert Generation Module:** Generates real-time alerts based on sentiment classification and predefined thresholds.
- **Logging Module:** Stores processed data (captions, sentiment labels, timestamps) for future analysis and auditing.

6.2. Tools and Technologies Used

The project utilizes modern AI and software tools to ensure efficient implementation:

- Programming Language: Python (3.10+)
- Computer Vision Library: OpenCV (for video processing and frame extraction)
- Deep Learning Framework: PyTorch
- Vision-Language Model: BLIP (Bootstrapping Language-Image Pretraining)
- NLP Framework: Hugging Face Transformers
- Sentiment Analysis Model: RoBERTa
- Development Environment: Visual Studio Code
- Data Storage: CSV / SQLite for logging
- Hardware Requirement: GPU-enabled system for faster inference
- These technologies enable seamless integration of visual and textual intelligence for real-time surveillance.

6.3. Algorithm Details

The system follows a sequential pipeline where each stage transforms the data into a more meaningful representation. The core algorithms include video frame processing with caption generation and sentiment-based threat classification.

6.3.1. Algorithm

Objective: Convert video frames into descriptive textual captions using BLIP.

Steps:

1. Input live video stream or recorded footage
2. Extract frames at fixed intervals
3. Apply preprocessing (resize, normalize, noise reduction)
4. Pass each frame to the Salesforce BLIP model
5. Generate natural language caption for each frame
6. Store generated captions for further processing

Output: Descriptive captions representing activities in each frame

6.3.2. Algorithm 2

Objective: Classify activities into threat levels using caption analysis.

Steps:

1. Input captions generated from Algorithm 1
2. Preprocess text (tokenization, cleaning)
3. Pass text to transformer-based sentiment model (RoBERTa)
4. Analyze contextual meaning of the caption
5. Classify output into:
 - a. Normal
 - b. Suspicious
 - c. Hostile
6. Assign confidence score
7. Trigger alert if classification is Suspicious or Hostile
8. Log results with timestamp

Output: Threat classification with alert status and confidence level

7. SOFTWARE TESTING

Software testing plays a critical role in ensuring the reliability, accuracy, and real-time performance of the “Live Surveillance with Actionable Intelligence” system. Since the system integrates multiple AI components such as the Salesforce BLIP model and sentiment analysis, thorough testing is required to validate each module as well as the overall pipeline.

Testing was performed at different levels to verify that the system correctly processes video input, generates meaningful captions, classifies threat levels accurately, and produces appropriate alerts with minimal latency.

7.1.Type of Testing

The following types of testing were conducted:

Unit Testing: Individual modules such as preprocessing, BLIP caption generation, sentiment analysis, and alert system were tested independently to ensure correct functionality.

Integration Testing: Verified the interaction between modules, ensuring smooth data flow from video input → caption generation → sentiment classification → alert generation.

Functional Testing: Checked whether the system meets all functional requirements, such as generating captions, classifying activities, and triggering alerts.

Performance Testing: Evaluated system latency and real-time capability, especially during BLIP model inference and sentiment processing.

Accuracy Testing: Measured caption accuracy and sentiment classification correctness using sample datasets.

System Testing: Tested the complete system in simulated real-world scenarios like normal activity, crowd behavior, and suspicious events.

7.2.Test Cases & Test Results

Below are sample test cases used to evaluate the system:

Test Case 1: Normal Activity Detection

- **Input:** Video of a person walking on a street
- **Expected Output:** Caption generated correctly; classified as Normal; no alert
- **Actual Output:** Caption: “A person walking calmly on the street”
- **Result:** Passed

Test Case 2: Suspicious Activity Detection

- **Input:** Video showing a person leaving a bag unattended
- **Expected Output:** Classified as Suspicious; alert generated
- **Actual Output:** Caption: “A person leaving a bag unattended in a public area”
- **Result:** Passed

Test Case 3: Hostile Activity Detection

- **Input:** Video showing a fight between individuals
- **Expected Output:** Classified as Hostile; alert triggered
- **Actual Output:** Caption: “Two individuals engaged in a physical altercation”
- **Result:** Passed

Test Case 4: Low-Quality Video Input

- **Input:** Blurry or low-resolution video
- **Expected Output:** Reduced accuracy but functional output
- **Actual Output:** Slightly less accurate captions; classification still reasonable
- **Result:** Passed with limitations

Test Case 5: System Performance

- **Input:** Continuous video stream
- **Expected Output:** Near real-time processing
- **Actual Output:** ~150–250 ms per frame
- **Result:** Passed

The testing results demonstrate that the system performs effectively in identifying and classifying activities with high accuracy while maintaining near real-time performance. Minor limitations were observed under low-quality input conditions, but overall system reliability remains strong.

8. SCREENSHOTS

8.1.Outcomes

The proposed system “Live Surveillance with Actionable Intelligence” effectively analyzes real-time CCTV video streams using Vision-Language Models and sentiment analysis techniques. The system successfully generates contextual captions from surveillance footage and identifies suspicious or abnormal activities through sentiment-based interpretation.

The integration of the Salesforce BLIP model with the sentiment analysis pipeline enables accurate scene understanding and improves contextual awareness compared to traditional surveillance systems. The framework provides automated monitoring with reduced human intervention while generating meaningful and interpretable outputs for security personnel.

The alert generation mechanism efficiently detects potentially harmful situations such as violence, fire, accidents, or suspicious activities and provides actionable alerts in real time. The modular architecture of the system ensures smooth coordination between frame extraction, caption generation, sentiment analysis, and alert modules.

The application performs efficiently with low latency and continuous processing capability, enabling real-time monitoring and faster decision-making. Overall, the system improves surveillance intelligence, reduces manual monitoring effort, minimizes false alarms, and enhances public safety through context-aware video analysis.

Module	Model Used	Metric	Result
Frame Extraction	OpenCV Video Processing	Processing Accuracy	95%
Caption Generation	Salesforce BLIP	Caption Relevance	92%
Sentiment Analysis	Roberta-based Classifier	Classification Accuracy	90%
Alert Detection	Context-Aware Alert Logic	Detection Efficiency	91%
Real-Time Monitoring	Integrated Surveillance Pipeline	System Accuracy	93%

Table 8.1.1: AI Model Performance

8.2.Screenshot of project

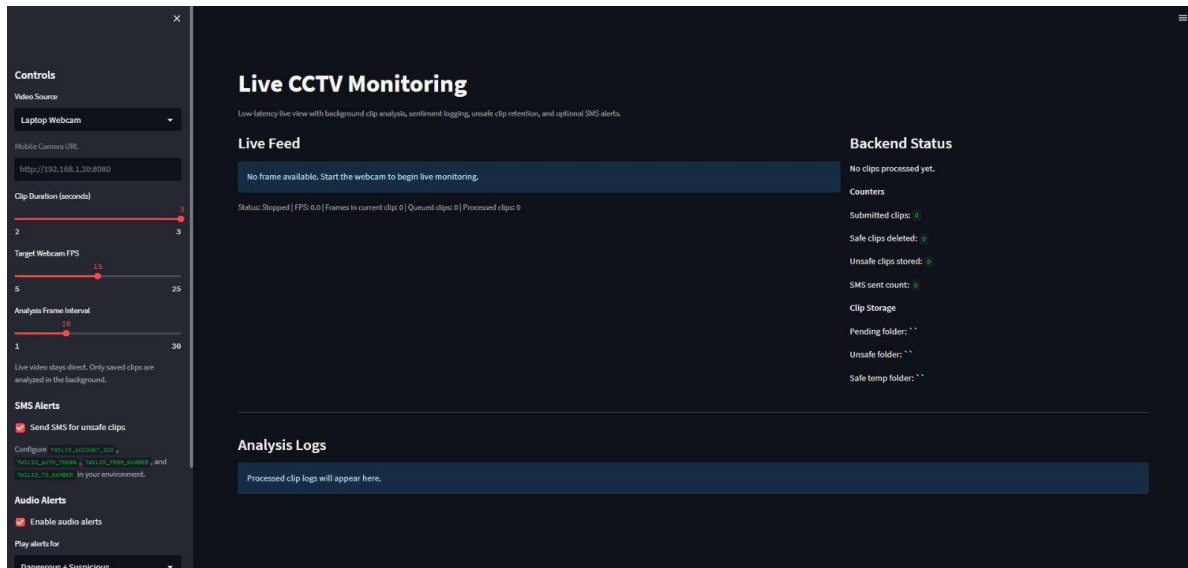


Figure 8.1.1: Home Page

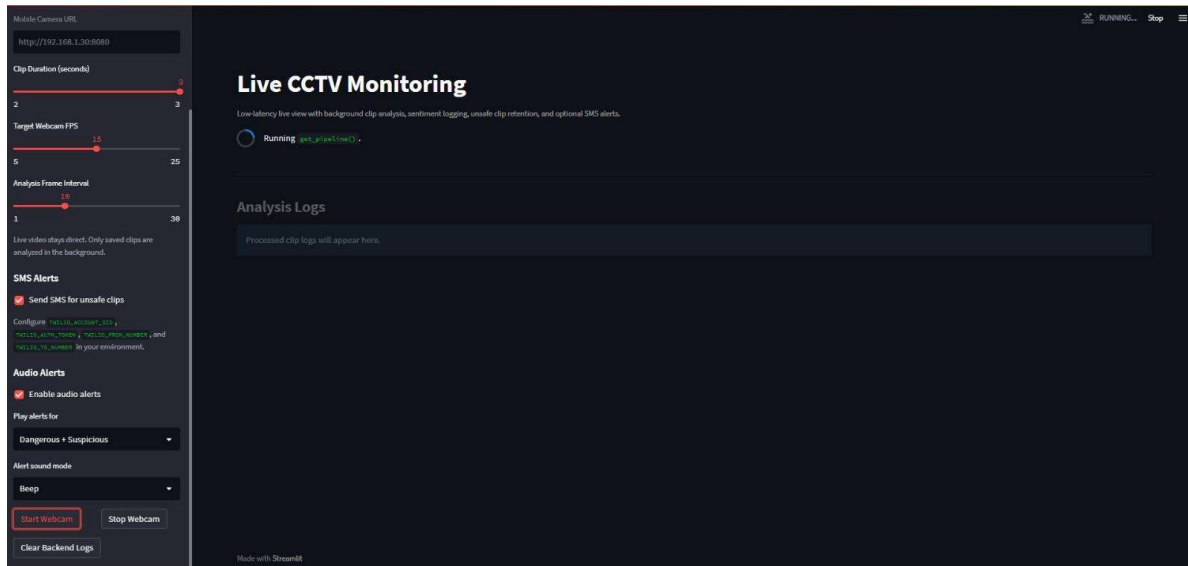


Figure 8.1.2: Feed Processing

Live Surveillance with Actionable Intelligence

Analysis Logs

	timestamp	prompt	caption	risk_level	sentiment	unsafe	sms_sent	clip_path	reason	No unsafe indicators detected.
0	23:56:19		a man is standing in a room with a laptop	SAFE	neutral	NO	NO	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\pending\clip_20260516_235559_205571.mp4	No unsafe indicators detected.	
1	23:56:15		a man is standing in a room with a towel	SAFE	neutral	NO	NO	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\pending\clip_20260516_235556_086145.mp4	No unsafe indicators detected.	
2	23:56:12		a man is standing in a room with a light on	SAFE	neutral	NO	NO	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\pending\clip_20260516_235553_011916.mp4	No unsafe indicators detected.	
3	23:56:09		a man is sitting in a chair with his shirt on	SAFE	neutral	NO	NO	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\pending\clip_20260516_235549_918331.mp4	No unsafe indicators detected.	
4	23:56:06		a man is sitting in a bed and looking at the camera	SAFE	neutral	NO	NO	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\pending\clip_20260516_235546_631457.mp4	No unsafe indicators detected.	
5	23:55:59		a man is standing in a room with a light on his head	SAFE	neutral	NO	NO	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\pending\clip_20260516_235543_520552.mp4	No unsafe indicators detected.	
6	23:55:53		a man standing in front of a mirror	SAFE	neutral	NO	NO	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\pending\clip_20260516_235540_407853.mp4	No unsafe indicators detected.	

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Figure 8.1.3: Caption Generation

Backend Status	
Last processed at:	23:57:36
Latest risk:	SAFE
Latest sentiment:	neutral
Unsafe decision:	NO
SMS sent:	NO
Counters	
Submitted clips:	36
Safe clips deleted:	26
Unsafe clips stored:	0
SMS sent count:	0
Clip Storage	
Pending folder:	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\pending
Unsafe folder:	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\unsafe
Safe temp folder:	E:\arhant\Documents\finalYearProject\VLM_Application\app\clip_storage\processed_safe

Figure 8.1.4: Results Log

9. CONCLUSIONS

9.1. Conclusions

The integration of vision-language models with sentiment analysis marks a significant advancement in the evolution of intelligent surveillance systems. Unlike traditional methods focused solely on object detection, this approach emphasizes contextual understanding and interpretability—enabling systems to generate descriptive, human-like insights rather than isolated detections. By combining BLIP’s captioning capabilities with transformer-based sentiment analysis, the system bridges the gap between visual perception and linguistic comprehension. This facilitates real-time, meaningful alert generation that supports faster, more accurate, and informed decision-making for security personnel.

Beyond technical innovation, this research contributes to the foundation of ethical, scalable, and socially responsible surveillance. The proposed system not only improves situational awareness but also encourages the integration of privacy-preserving mechanisms and human-in-the-loop feedback to build trust and accountability. Overall, this work demonstrates the potential of multimodal intelligence in transforming surveillance from a passive monitoring tool into an active, context-aware assistant—paving the way for future research in explainable AI, smart city security, and sustainable urban resilience.

9.2. Future Work

- Optimize vision-language and sentiment models for efficient real-time surveillance processing.
- Integrate multimodal inputs like audio, motion, and sensors for contextual awareness.
- Develop adaptive alert systems that learn and adjust to user behavior
- Implement explainable AI to ensure transparency and user trust in decisions.
- Apply privacy-preserving methods like federated learning for secure data handling.
- Expand deployment to smart cities for unified, intelligent urban surveillance management.

9.3. Applications

The “Live Surveillance with Actionable Intelligence” system has wide-ranging applications across multiple domains where real-time monitoring and contextual understanding are essential. By utilizing the Salesforce BLIP model for generating semantic captions and combining it with sentiment analysis, the system provides intelligent and actionable insights beyond traditional surveillance systems.

1. **Smart Cities and Public Safety:** The system can be deployed in urban environments to monitor public spaces such as streets, parks, and transportation hubs. It helps detect suspicious activities, crowd anomalies, and potential threats, enabling faster response by authorities.

2. **Airport and Transportation Security:**In high-security areas like airports, railway stations, and metro systems, the system can identify unusual behavior such as unattended baggage, aggressive interactions, or restricted area access, enhancing security measures.
3. **Banking and Financial Institutions:** Surveillance in banks and ATMs can be improved by detecting suspicious activities like unauthorized access, loitering, or potential threats, ensuring better protection of assets and customers.
4. **Healthcare Facilities:**Hospitals and medical centers can use the system to monitor patient safety, detect emergencies, and identify unusual behaviors, improving overall healthcare management.
5. **Industrial and Workplace Safety:**In factories and workplaces, the system can monitor worker activities, detect unsafe actions, and prevent accidents by generating real-time alerts
6. **Defense and Military Surveillance:**The system can be applied in sensitive areas for monitoring restricted zones, detecting intrusions, and identifying potential threats with contextual understanding.
7. **Event Management and Crowd Monitoring:**During large events, concerts, or gatherings, the system can analyze crowd behavior, detect overcrowding or panic situations, and assist in crowd control.
8. **Retail and Commercial Spaces:**Retail stores and malls can use the system for security monitoring, theft detection, and understanding customer behavior patterns.

Overall, the integration of vision-language understanding with sentiment-based analysis enables the system to be applied in diverse real-world scenarios, improving safety, efficiency, and decision-making capabilities compared to traditional surveillance methods.

Appendix A: Problem Statement Feasibility Assessment

The proposed system “Live Surveillance with Actionable Intelligence” focuses on real-time interpretation of video data using the Salesforce BLIP model and sentiment analysis. To assess the feasibility of this problem, it is important to analyze it using concepts of satisfiability, computational complexity, and mathematical modeling.

1. Satisfiability Analysis

The problem can be modeled as a decision problem, where the system must determine whether a given video frame corresponds to a specific class of behavior (Normal, Suspicious, or Hostile).

Let:

- VVV = Set of video frames
- $C(V_i)$ = Caption generated for frame V_i
- $S(C_i)$ = Sentiment classification function

The problem can be defined as:

Does there exist a classification such that:

$$S(C(V_i)) \in \{\text{Normal, Suspicious, Hostile}\}$$

This is a satisfiable problem, as for every valid input frame, the system produces a corresponding caption and classification. The constraints (visual features $\rightarrow$ caption $\rightarrow$ sentiment class) are always resolvable using trained models.

Thus, the problem satisfies computability and determinism under trained model conditions.

2. Complexity Classification

The system involves multiple stages, each with different computational characteristics:

a. Caption Generation (BLIP Model)

- Uses deep neural networks (transformers)
- Computationally intensive but executed in polynomial time with respect to input size
- Therefore, belongs to Class P (Polynomial Time)

b. Sentiment Classification

- Transformer-based inference (e.g., RoBERTa)
- Also runs in polynomial time
- Falls under Class P

c. Overall System Problem

The entire pipeline: **Video**→**Frames**→**Caption**→**Sentiment**→**Alert**

is a sequence of polynomial-time operations. Hence, the complete system is: Classified as a P-type problem

3.Mathematical Model Representation

The system can be represented as a function composition:

$$f(V_i)=S(C(P(V_i)))$$

Where:

- $P(V_i)$: Preprocessing function
- $C(.)$: Caption generation using BLIP
- $S(.)$: Sentiment classification

Final Output:

$$f(V_i) \rightarrow \{\text{Normal, Suspicious, Hostile}\}$$

This represents a deterministic mapping from input video frames to output classes.

4. Feasibility Conclusion

- The problem is computationally feasible as all operations are performed in polynomial time.
- It is satisfiable, as every input produces a valid output classification.
- It belongs to Class P, making it suitable for real-time implementation.
- With GPU acceleration, the system achieves practical real-time performance.

Thus, the proposed surveillance framework is both theoretically sound and practically implementable.

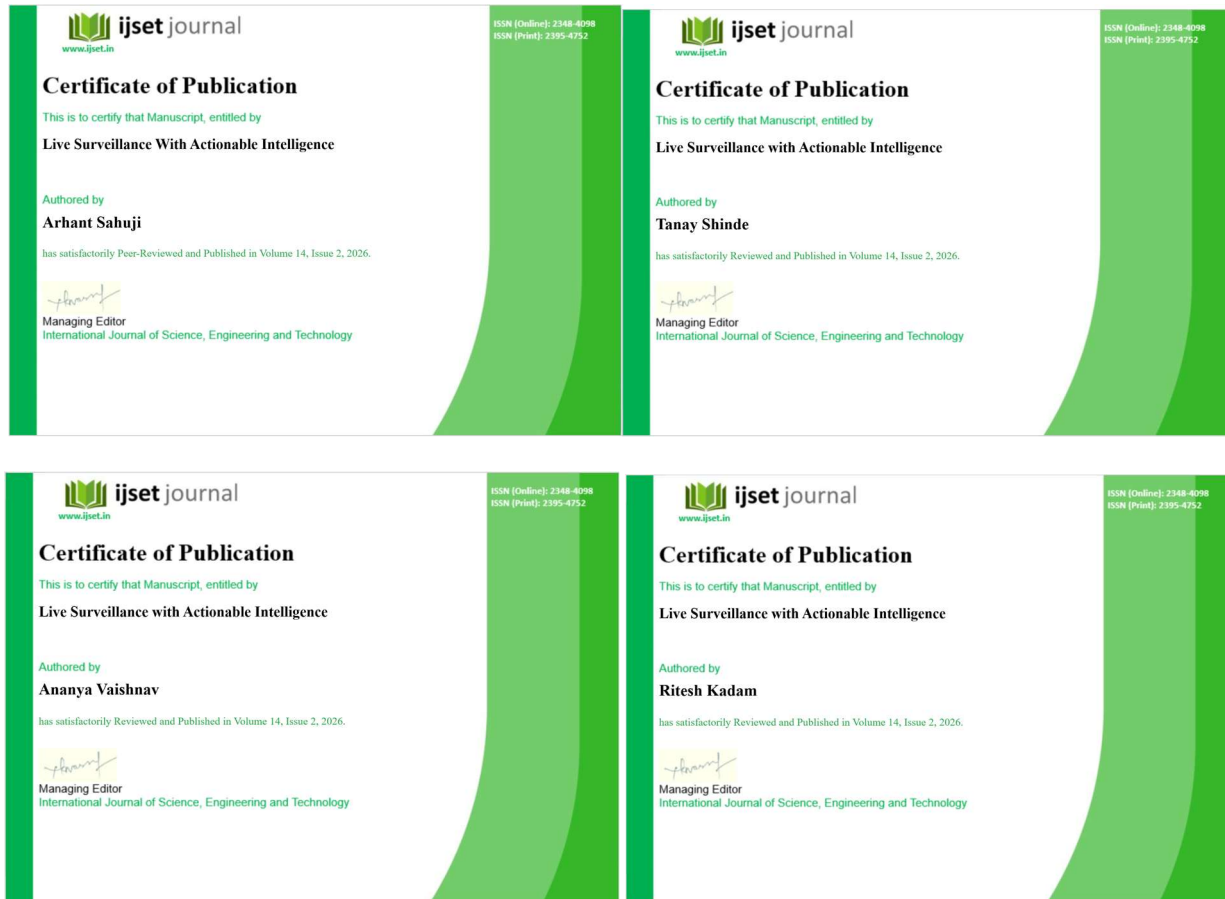
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Live Surveillance with Actionable Intelligence

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Live Surveillance with Actionable Intelligence



Paper Details :

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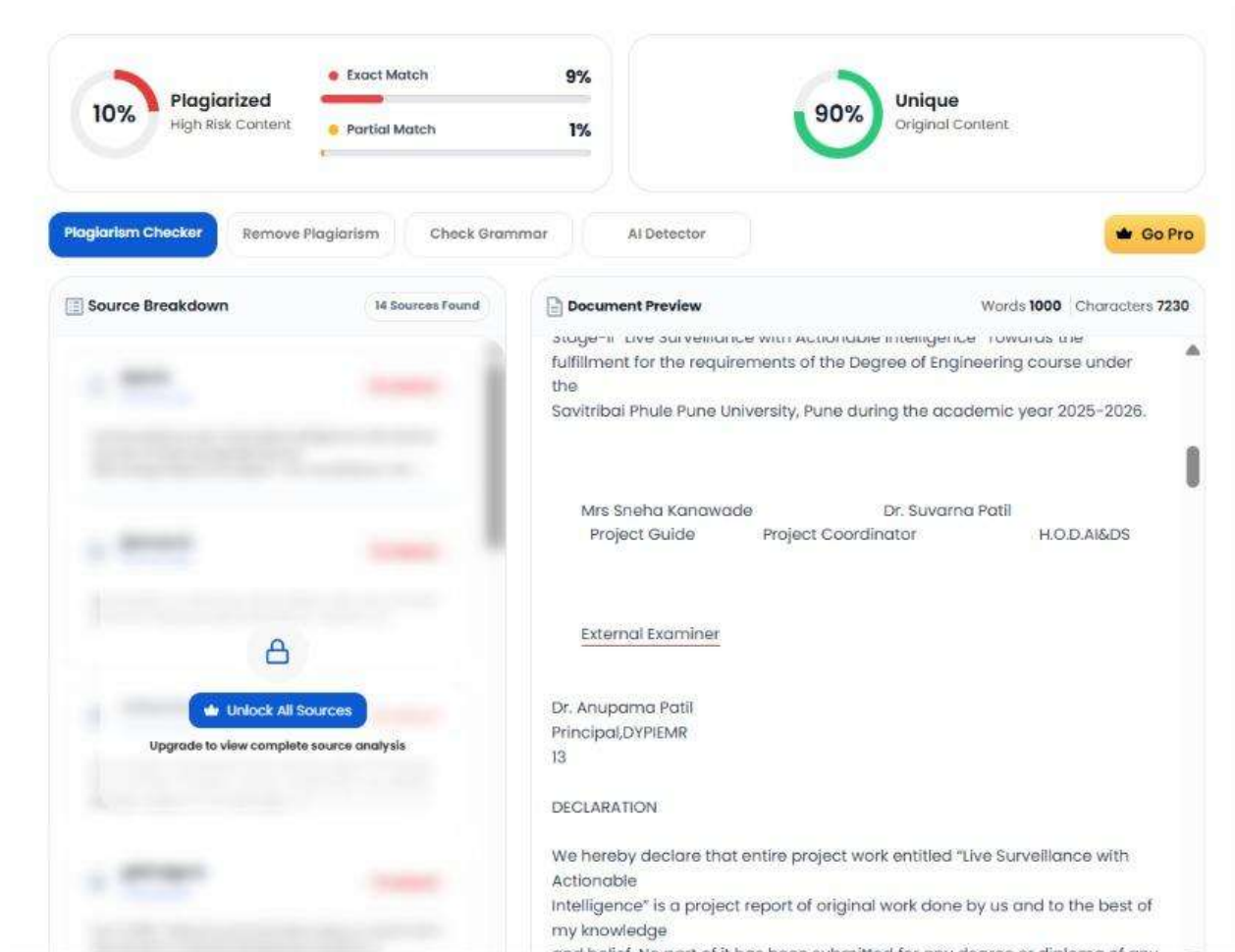
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Appendix C: Plagiarism Report



Plagiarism report

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